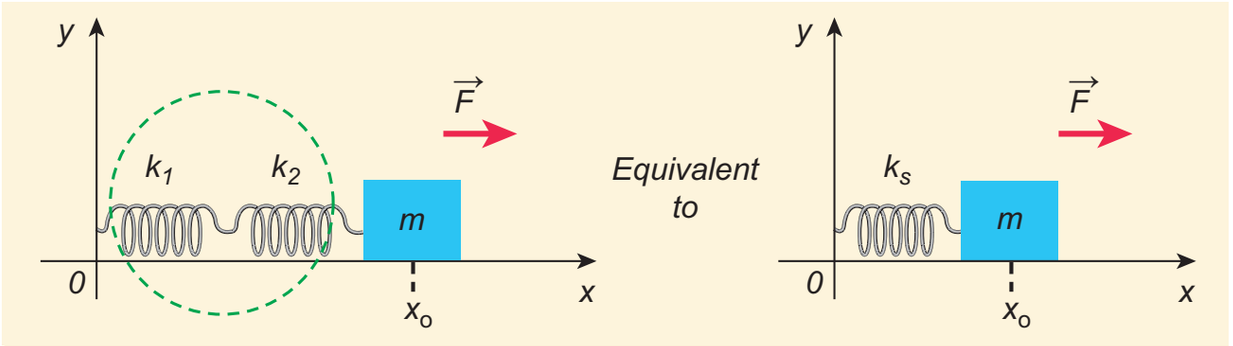


Figure 10.18 Effective spring constant in series connection

$$F = -k_s(x_1 + x_2) \Rightarrow x_1 + x_2 = -\frac{F}{k_s} \quad (10.38)$$

For springs in series connection

$$-k_1x_1 = -k_2x_2 = F$$

$$\Rightarrow x_1 = -\frac{F}{k_1} \text{ and } x_2 = -\frac{F}{k_2} \quad (10.39)$$

Therefore, substituting equation (10.39) in equation (10.38), the *effective spring constant* can be calculated as

$$-\frac{F}{k_1} - \frac{F}{k_2} = -\frac{F}{k_s}$$

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2}$$

Or

$$k_s = \frac{k_1k_2}{k_1 + k_2} \text{ Nm}^{-1} \quad (10.40)$$

Suppose we have "*n*" springs connected in series, the effective spring constant in series is

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \ldots + \frac{1}{k_n} = \sum_{i=1}^{n} \frac{1}{k_i} \quad (10.41)$$

If all spring constants are identical i.e., $k_1 = k_2 = \ldots = k_n = k$ then

$$\frac{1}{k_s} = \frac{n}{k} \Rightarrow k_s = \frac{k}{n} \quad (10.42)$$

This means that the effective spring constant reduces by the factor "*n*". Hence, for springs in series connection, the effective spring constant is lesser than the individual spring constants.

From equation (10.39), we have,

$$k_1x_1 = k_2x_2$$

Then the ratio of compressed distance or elongated distance x_1 and x_2 is

$$\frac{x_2}{x_1} = \frac{k_1}{k_2} \quad (10.43)$$

The elastic potential energy stored in first and second springs are $U_1 = \frac{1}{2}k_1x_1^2$ and $U_2 = \frac{1}{2}k_2x_2^2$ respectively. Then, their ratio is

$$\frac{U_1}{U_2} = \frac{\frac{1}{2}k_1x_1^2}{\frac{1}{2}k_2x_2^2} = \frac{k_1}{k_2}\left(\frac{x_1}{x_2}\right)^2 = \frac{k_2}{k_1} \quad (10.44)$$

Note

The reciprocal of stiffness constant is called flexibility constant or compliance, denoted by C. It is measured in m N^{-1}

If n springs are connected in series :

$$\text{net compliance } C_s = \sum_{i=1}^{n} C_i$$

If n springs are connected in parallel :

$$\frac{1}{C_p} = \sum_{i=1}^{n} \frac{1}{C_i}$$

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